

Rates

Name: _____

Work on these problems with those around you. We will be submitting this at the end of class for credit instead of a quiz this week.

1. A bicycle is traveling at 15 miles per hour. How many feet will it cover in 20 seconds?
2. Suppose you have \$600 billion dollars in \$100 dolar bills, and you are counting them out 1 per second (or maybe you are paying someone else to count them for you), how long will it take to count them?

3. You and your freinds wish to share a large pizza (12 inches in diameter). After placing the order, the manager comes out and explains to you that they are all out of large pizzas, and that it will be a long wait until another is ready, but he can offer you two medium pizzas (8 inches in diameter) now. Should you take the medium pizzas, or wait for the large? (It may be helpful to recall that the area of a circle is given by $A = \pi r^2$, where r is the radius of the circle, also assume that all pizzas are perfect circles).
4. Suppose that you need 48 fifty-pound bags of sand to fill a sandbox that is 8 feet wide, 8 feet long, and 1 foot deep. How many fifty-pound bags of sand will you need to fill a sandbox that is 4 feet wide, 4 feet long, and 1 foot deep?
5. A recipe for zucchini muffins states that it yields 12 muffins, with 250 calories per muffin. You instead decide to make mini-muffins, and the recipe yields 20 muffins. If you eat 4, how many calories will you consume?